

TITLE PAGE

- **Problem Statement ID** - 25217
- **Problem Statement Title** - Cognitive Opportunistic LEO-PNT Receiver System
- **Theme** - Space Technology
- **PS Category** - Hardware
- **Team ID** -72891
- **Team Name** –Helix House



Problem:

GNSS signals are vulnerable to jamming, spoofing, and blocking. In battlefields and tough terrains, this leaves soldiers without reliable navigation, **putting missions and lives at risk**

Our Idea :

ZyraNav is an AI-powered device with a three-layer approach ensuring reliable navigation when GPS fails. It uses simulated LEO signals to train adaptive algorithms for robust operation in any scenario.

Proposed Solution :

- When **GNSS fails**, the system seamlessly falls back to **3-layer architecture** — GNSS (primary), LEO SoO (secondary), and inertial navigation system-INS (tertiary).
- A **smart simulation framework** generates synthetic LEO SoO signals from GNSS data to train and validate cognitive algorithms.
- An **AI-enabled cognitive algorithm** and a **Low-cost handheld device** with smart power management for 24-hour continuous operation.

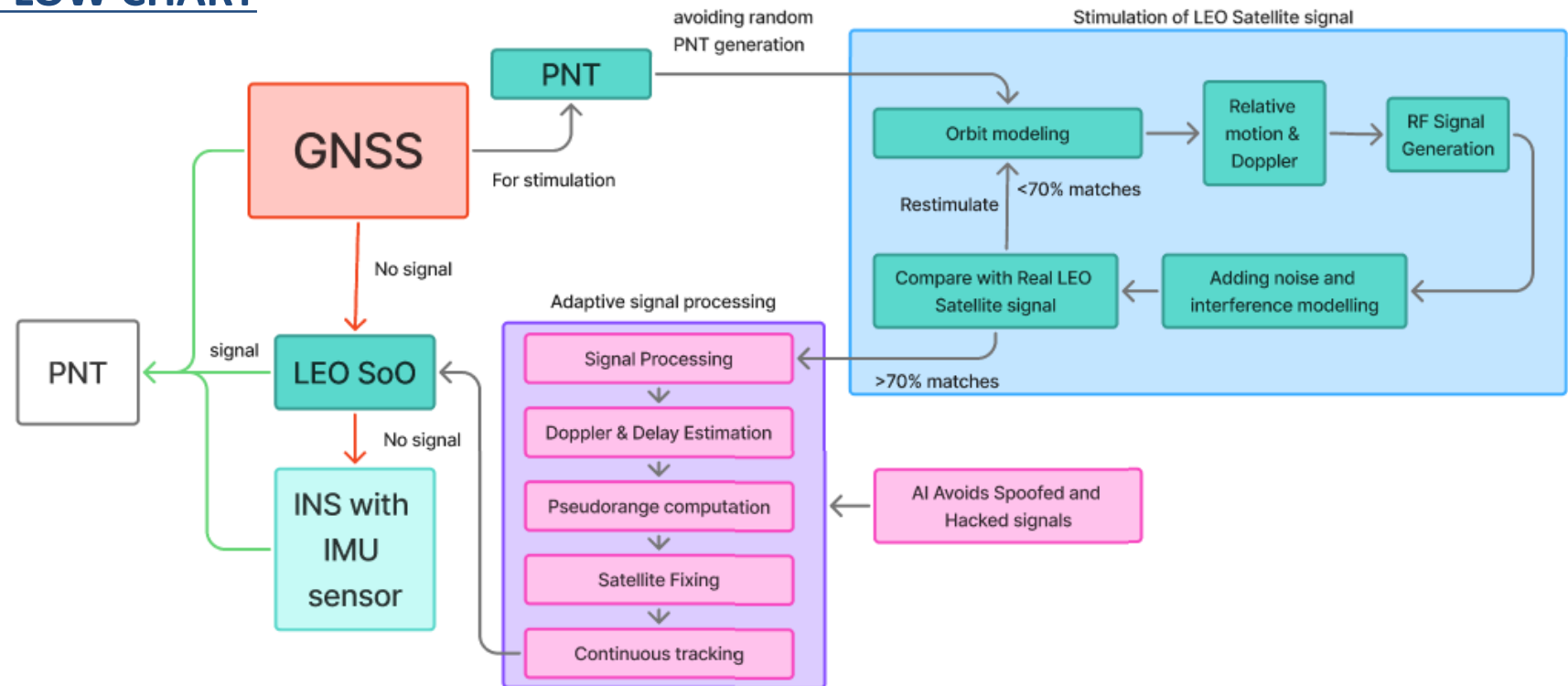
**Innovation/Uniqueness:**

- **India's first 3-layer navigation system** to get PNT
- **Transforms non-PNT LEO satellite signals** into usable PNT data through adaptive signal conversion.
- **AI-powered adaptive cognitive algorithm** detects GNSS failure, switches between LEO and INS, and performs adaptive signal processing for high accuracy.
- **Lab-based Smart Signal Simulation Framework** lowers development cost and trains the cognitive model for real-world robustness.

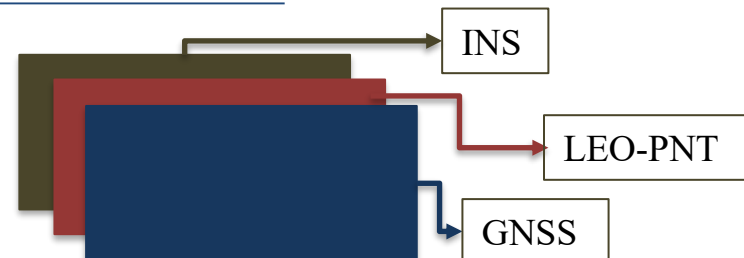
Hardware & Software

- **GNSS Receiver Module (u-box ZED-F9P):** Primary PNT source for GPS, NavIC, and multi-constellation tracking.
- **IMU (ICM20948):** Backup inertial navigation (INS)
- **Raspberry Pi 4 (4GB):** Runs LEO signal simulation, AI algorithms, and sensor fusion.
- **LimeSDR (optional):** Captures real LEO signals or uses public datasets.
- **Python / NumPy / SciPy:** Simulate, validate, and compare LEO SoO signals.
- **TensorFlow Lite / PyTorch Mobile:** AI for detecting spoofed/jammed signals and selecting the most reliable sources.
- **Battery (Li-ion 25,000 mAh):** 24-hour continuous operation.
- **3.5-inch Color TFT LCD:** Displays maps, navigation data, and system status.

FLOW CHART



3 LAYER APPORACH



GitHub link:

<https://github.com/sangsaist/ZyraNav.git>

YouTube video : [Link](#)

Above 40% of the prototype is completed

Feasibility

- Lab based **simulated LEO SoO signals**, avoiding early hardware dependency.
- Converts GNSS PNT into LEO SoO signals for testing.
- **Validation with official datasets** Ensures accurate simulation by comparing with real GNSS data.
- **Safe testing environment:** No RF transmission in air , all tests done internally.
- **Algorithm flexibility:** Controlled simulation allows the AI algorithm to be trained for **any condition** before real-time deployment.

Challenges

- **Fast- Moving LEO satellites** cause frequent Doppler changes.
- Commercial LEO signals vary in **modulation and timing**.
- Real-time cognitive processing **consumes high power**.
- GNSS and SoO signals can be **tampered with in contested areas**.

Commercial feasibility

- **Market Validation:** Growing interest in LEO-based navigation shows demand for resilient PNT solutions.
- **Technical Readiness:** ~85% feasible using **simulated LEO signals compared with publicly available datasets**.
- **Timeline:** 5 days prototype using **stimulated LEO signals and comparison with real LEO Signal data**.
- **Success Likelihood:** High (~90%) if simulation accurately models LEO SoO and AI reliably selects signals;

Strategy:

- **Adaptive frequency** estimation and Doppler compensation using **Kalman filters**.
- **AI-based pattern recognition** trained on open-source LEO RF datasets to decode generic parameters.
- **Edge-optimized AI models** on Raspberry Pi, dynamic task scheduling, and energy-efficient algorithms.
- **Fuse GNSS, simulated LEO SoO, and INS data;** implement anomaly detection to reject compromised signals.

Direct Impact on Target Users

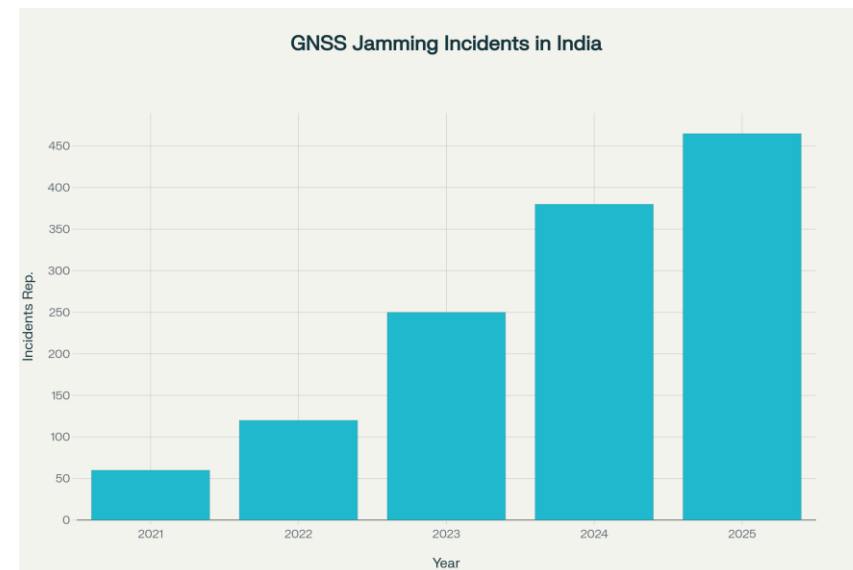
- **Defense & Security Forces:** Enables **resilient navigation** in GPS/GNSS-denied hostile environments (mountain borders, tunnels, ECM zones).
- **Disaster Response Teams (NDRF/SDRF):** Provides **~30% faster rescue operations**, improving coordination in collapsed or forested zones.
- **Rural & Remote India:** Ensures **accurate positioning** where GPS signals are weak or unavailable (forests, valleys, underground areas).
- **Transport & Logistics (Rail, Ports, Drones):** Ensures **reliable, uninterrupted routing** even under jamming or satellite shadowing.

Strategic Impact

- Offers **GPS-independent PNT**, highly **resilient to jamming, spoofing, and signal outages**.
- Strengthens India's autonomy in **defense, disaster management, rural navigation, and space technology**.
- **Impact:** Contributes to *Atmanirbhar Bharat* by establishing **indigenous LEO-PNT capability** for national security and infrastructure.

Economic & Strategic Benefits

- **Reduces GNSS licensing/import costs by 30%**, promoting indigenous receiver development.
- **₹4,750 crore** total addressable market by **2030** in defense, logistics, and disaster management sectors.
- Superior cost effectiveness over traditional anti-jamming GPS receivers through Ai-enabled cognitive efficiency.
- Encourages **public-private partnerships (PPP)** for scalable Indian navigation tech.



Gap & Problem Identification

- *Quote recent studies* documenting the rise in GNSS jamming/spoofing events and failures in Indian context. [link](#)
- Show comparative performance of traditional GNSS (GPS, NavIC) vs. LEO-PNT and INS in denied environments. [link](#)

Economic & Strategic Landscape

- Market forecasts: cite projected growth of India's LEO-PNT sector and expected impacts on defense, transportation, and PPP innovation 1. [link](#) 2. [link](#)

Literature Survey & Competitive Analysis

- Summarize latest global and Indian research on LEO-PNT market trends, strengths, and competitive technologies (Starlink, Iridium, NavIC, VyomIC). 1. [link](#) 2. [link](#) 3. [link](#)
- Present findings from peer-reviewed experiments comparing GNSS, LEO, and INS fallback capabilities in real-world urban, remote, and ECM scenarios. 1. [link](#) 2. [link](#)

Field Tests & Simulation Results

- Include results from your own simulation/bench tests (e.g., improvement in rescue operation time, accuracy rates in tunnels or shadow zones). [link](#)
- Reference international proof-of-concept studies showing LEO-aided INS tested on vehicles or UAVs achieving meter-level accuracy where GNSS failed [link](#)

Technology Benchmarking

- Compare accuracy, jamming resilience, and coverage of GNSS with new LEO systems (e.g., meter-level positioning under jamming from LEO vs. multi-meter GNSS inaccuracies). [link](#)
- Highlight integration of AI-driven cognitive algorithms and multi-sensor fusion for spoofing defense and PNT integrity. [link](#)

Policy & Ecosystem Analysis

- Mention government and military initiatives for anti-spoofing, indigenous navigation frameworks, and Atmanirbhar Bharat alignments. [link](#)